

Table 1: Every-bar (refit = 1) exogenous-bucket battery vs. the HAR baseline. Train window 1000 trading days; fixed COVID-inclusive evaluation window **Jan. 2018 – May 2025** ($n = 74,934$ thirty-minute bars; 1,843 trading days; identical rows for every cell); best penalty per bucket selected by QLIKE; HAR block unpenalized (OLS via Frisch–Waugh–Lovell). DM = Diebold–Mariano test on per-bar QLIKE loss vs. HAR-only, reported as (t, p) : the DM statistic is the t -ratio of the mean loss differential to its Newey–West HAC standard error (Harvey–Leybourne–Newbold small-sample correction), with two-sided p -value. The 90% model confidence set contains ALL buckets alone; every other bucket is eliminated. Raw-variance MSE and OOS- R^2 are dominated by a small number of COVID bars; QLIKE, HMSE and MAE are the level-robust rankers.

exogenous bucket	penalty	QLIKE	MAE	MSE	OOS- R^2	MZ- β	MZ- R^2	DM vs. HAR (t, p)
ALL buckets	ridge $\alpha=30$	0.14265	1.213×10^{-6}	4.571×10^{-11}	0.6969	0.902	0.7058	$-13.0, 1.2 \times 10^{-38}$
implied_vol (VIX/VVIX)	ridge $\alpha=1$	0.14669	1.236×10^{-6}	4.558×10^{-11}	0.6978	0.907	0.7057	$-8.7, 2.3 \times 10^{-18}$
moments	ridge $\alpha=1000$	0.14678	1.246×10^{-6}	4.624×10^{-11}	0.6935	0.879	0.7074	$-7.5, 4.6 \times 10^{-14}$
liquidity	ridge $\alpha=1$	0.14679	1.250×10^{-6}	4.599×10^{-11}	0.6951	0.892	0.7060	$-10.8, 3.8 \times 10^{-27}$
sentiment	enet $\alpha=10^{-3}, \ell_1=0.6$	0.14726	1.250×10^{-6}	4.484×10^{-11}	0.7027	0.901	0.7120	$-8.2, 2.9 \times 10^{-16}$
market_vw	enet $\alpha=10^{-3}, \ell_1=0.2$	0.14744	1.253×10^{-6}	4.558×10^{-11}	0.6978	0.891	0.7091	$-8.7, 4.5 \times 10^{-18}$
market_ew	ridge $\alpha=1000$	0.14758	1.257×10^{-6}	4.593×10^{-11}	0.6955	0.889	0.7071	$-9.3, 1.0 \times 10^{-20}$
vol_demand	enet $\alpha=10^{-3}, \ell_1=0.6$	0.14822	1.259×10^{-6}	4.553×10^{-11}	0.6982	0.900	0.7075	$-6.2, 5.0 \times 10^{-10}$
<i>HAR-only (baseline)</i>	none (OLS)	0.14863	1.262×10^{-6}	4.545×10^{-11}	0.6987	0.898	0.7084	(baseline)

Table 2: Intra-bucket bundle-combination factorial (2^8 over the stage-1 surviving bundles; a bundle = one variable’s six-lag moving-average ladder plus its availability indicator). Model: ridge $\alpha = 1$ on the square-root target with the RV-HAR ladder and intraday hour dummies always included; every-bar refit. Primary window $tw = 48,000$ bars (1,000 trading days): evaluation **Oct. 9, 2008 – Apr. 30, 2024** ($n = 198,059$ bars). Replicate window $tw = 144,000$ (3,000 days): evaluation **Apr. 25, 2016 – Apr. 30, 2024** ($n = 102,059$ bars); its QLIKE level is not comparable across windows (different evaluation spans) — the ranking and DM are. DM (t -ratio, two-sided p ; as in Table 1) is vs. the HAR-baseline cell of the same window. QLIKE levels here are not comparable to Table 1: this panel is the raw research build (no diurnal adjustment or rank transform; overnight bars included), so the bundle gains include intraday seasonality that Table 1’s cache removes by construction — within-table rankings and DM are the deliverable. Exact factorial main effects on QLIKE, mean(bundle on) – mean(bundle off) over all 2^8 cells ($tw48k/tw144k$): absret $-0.199/-0.257$; volume $-0.108/-0.063$; vvix $-0.023/-0.014$; sentcount $-0.011/+0.003$; vdem_all $-0.010/+0.000$; attention $-0.010/+0.002$; vdem_spx $-0.009/+0.001$; bipow $-0.000/-0.000$. Largest first-order (two-way) interaction effects, same convention (positive = partial redundancy: the joint gain is less than additive): absret×volume $+0.094/+0.064$; absret×vvix $+0.020/+0.010$; volume×vvix $+0.014/-0.005$.

bundle combination	$tw = 48,000$ (2008–2024)					$tw = 144,000$ (2016–2024)	
	QLIKE	MAE	OOS- R^2	MZ- R^2	DM vs. HAR (t, p)	QLIKE	DM
<i>Main effects: the eight single-bundle cells (one bundle added to HAR), ranked by QLIKE</i>							
absret	0.38950	1.930×10^{-6}	0.3742	0.4592	$-37.3, < 10^{-300}$	0.38686	-58.8
volume	0.50567	2.561×10^{-6}	0.2159	0.2408	$-31.3, 1.6 \times 10^{-214}$	0.57475	-44.9
vvix	0.68515	2.912×10^{-6}	0.1257	0.1731	$-12.2, 2.1 \times 10^{-34}$	0.66325	-23.9
sentcount	0.71939	3.001×10^{-6}	0.0900	0.1099	$-11.9, 1.8 \times 10^{-32}$	0.69639	-7.1
vdem_all	0.72088	2.948×10^{-6}	0.0835	0.1117	$-14.7, 6.2 \times 10^{-49}$	0.68742	-8.4
attention	0.72629	3.012×10^{-6}	0.0903	0.1111	$-10.0, 2.0 \times 10^{-23}$	0.68812	-8.4
vdem_spx	0.73552	3.009×10^{-6}	0.0817	0.1059	$-12.6, 1.3 \times 10^{-36}$	0.67324	-13.0
bipow	0.78314	3.106×10^{-6}	0.0763	0.1004	$-6.2, 4.3 \times 10^{-10}$	0.72567	-2.1
<i>Selected combination cells: best by QLIKE; best-six of the P&L factorial; the full set</i>							
absret, volume, vdem_all, bipow	0.34598	1.862×10^{-6}	0.3629	0.4291	$-40.9, < 10^{-300}$	0.36599	-61.1
absret, volume, vdem_all, vvix, sentcount, attention	0.35237	1.862×10^{-6}	0.3771	0.4400	$-39.3, < 10^{-300}$	0.37061	-58.5
all eight bundles	0.35156	1.862×10^{-6}	0.3774	0.4399	$-39.4, < 10^{-300}$	0.37023	-58.1
<i>HAR ladder + hour dummies (baseline)</i>	0.78614	3.109×10^{-6}	0.0758	0.1002	(baseline)	0.72675	(baseline)

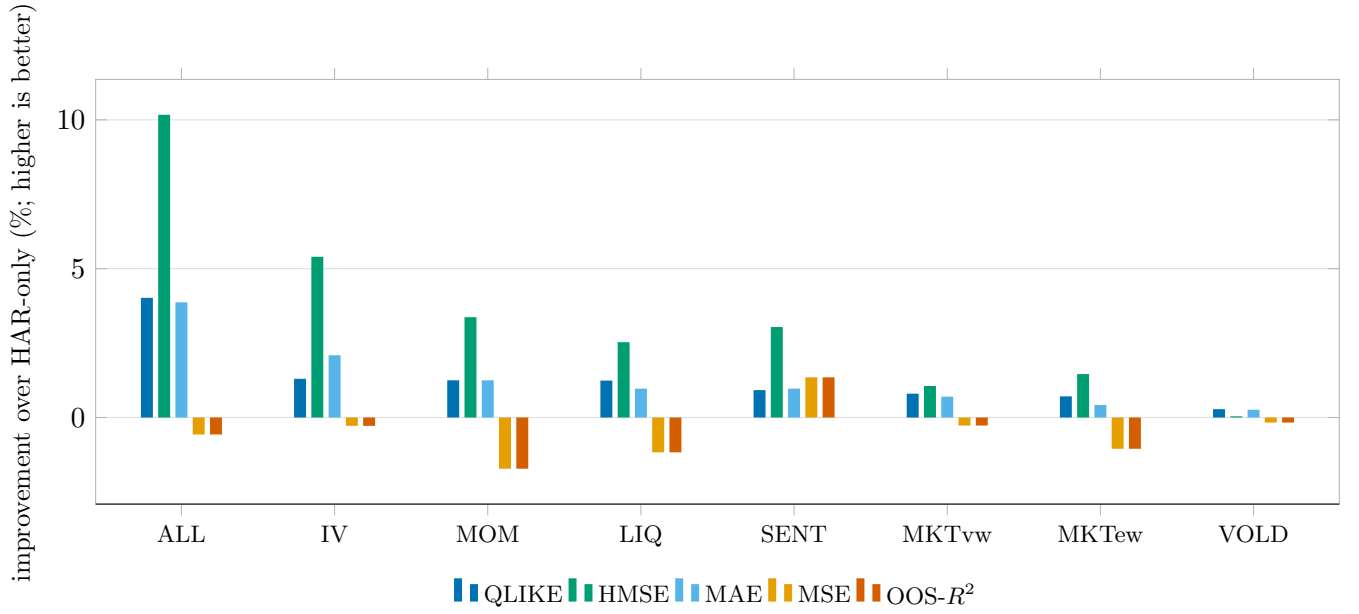


Figure 1: Every metric of Table 1 mapped to one comparable scale: fractional improvement over HAR-only, sign-oriented so that larger is better (the $OOS-R^2$ improvement is scaled by $1 - R^2_{HAR}$). The level-robust metrics (QLIKE, HMSE, MAE) agree — the full bucket set wins decisively (DM $p = 1.2 \times 10^{-38}$) — while raw-variance MSE and $OOS-R^2$ collapse toward zero or reverse sign, reflecting COVID-bar domination rather than forecast skill.